nature portfolio

Peer Review File

A single-dose circular RNA vaccine prevents Zika virus infection without enhancing dengue severity in mice

Corresponding Author: Professor Liqiang Feng

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
This manuscript by Liu and colleagues reports a preclinical study into a circular RNA vaccine to prevent Zika virus (ZIKV) infection. The authors compared 3 different constructs of circRNA to identify one that fused EDIII of ZIKV with a human IgG1 Fc fragment to form a EDIII dimer. They show that circRNA expressing EDIII-Fc produced superior immunogenicity compared to the other 2 constructs in adult female mice, and this superior immunogenicity translated to better protection against lethal ZIKV infection in pups from the vaccinated females. They also showed the the antibody response produced from vaccination cross-reacted minimally with DENVs and did not enhance DENV infection to levels seen with antisera against ZIKV. Finally, they showed that with additional modification, such as the use of IRES of iHRV-B3, the vaccine had the potential of being a single dose application with even better protection compared to their original construct. They concluded that their findings set a foundation for developing circRNA flavivirus vaccines.

This is an interesting study although there are several concerns that need to be addressed. These are:

1. Congenital Zika syndrome (CZS) is caused by ZIKV infection in expectant mothers, where transplacental transmission to the developing fetus dysregulates normal CNS development. In this study, ZIKV is inoculated to pups from vaccinated dams. Although I do not think the study design makes the findings less robust, a discussion on the differences in infection route between the experimental approach and real-world infection, along with their implication on vaccine-mediated protection against CZS should be clearly discussed.

2. Although the authors argued for the merits of circRNA, the dose used in this study – 20 microgram – is high. Conventional COVID-19 nucleoside-modified linear mRNA vaccines were showed efficacy against lethal SARS-CoV-2 infection in mice at a dose of 5 micrograms. The higher dose would mean not only higher cost, it could also lead to more adverse events and increase the risk of severe adverse events. Moreover, there is no comparison between circRNA and linear, nucleoside modified mRNA that express the same 3 different EDIII constructs. Thus, while the authors argued for the merits of circRNA, there is no data to show that circRNA is a better system to deliver the EDIII-Fc transgene.

3. Lines 56-57. This statement is not true. There are Zika vaccine candidates that have completed phase I clinical trial with useful safety outcomes. Their development has been stalled by the global low incidence of Zika cases. Please correct this statement.

4. Lines 58-59. Zika vaccine development has not been impeded by the concern of ADE of DENV infection. Please see explanation in comment #3.

5. Line 74. It is not helpful to mix the pathology of dengue with that of Zika. Coagulopathy and vascular leakage is more pertinent to dengue than Zika.

6. Line 122. Description of results in this line and elsewhere should be described as objectively as possible. Please avoid superlatives, such as “greatly”.

7. Lines 144-147 and Figure 1K. Although the differences in CD8+ T cell response to the different vaccine constructs raises interesting questions on the expression or processing of the different EDIII constructs for MHC class I presentation, it raises

concern for EDIII-Fc as a human vaccine. Human CD8+ T cell epitopes are found mostly elsewhere in the ZIKV and flaviviral proteome. If the protection seen in subsequent experiments, particularly those on adult IFNAR-/- mice were to be mediated, even in parts, by CD8+ T cells, then the vaccine efficacy seen in this mouse model may over-estimate the efficacy that can be expected in humans. Indeed, this possibility is raised by the data in Figures 4h and 5k, where the neutralising antibody titers produced after vaccination were not at levels sufficient for sterilizing immunity, as evidenced by detectable viremia even for the most optimal construct. Control of infection at later time points could thus have been mediated by cellular immunity, given that we now know anamnestic T cell response is faster than B cell response (Painter et al, Nat Immunol 2023; 24:1711-1724).

8. Line 248 and 384-392. Please specify the strain of mouse-adapted DENV-2 used in this study. What was the parental strain of this mouse-adapted DENV-2? Moreover, it is puzzling that the authors generated this strain by inoculating C57Bl/6 mice subcutaneously and could then harvest the virus from the brains of these animals. The reference they cited (Orozco et al, J Gen Virol 2012) used AG129 mice and passaged the virus for 11 cycles and not just 3 passages. Is this correct?

9. Lines 250-260. These statements misrepresent the findings in Figures 4d, 4e and 4g. Even sera from vaccinated animals led mortality and viremia in this model. Moreover, p=0.9179 in the survival curve for EN(LNP) vaccinated mice vs uninfected controls is simply unbelievable given that over 50% of the mice died from infection.

10. Figure 4a and Lines 914-916. This set of data cannot be shown in such a way. The IgG titers were measured using different approaches. Those for ZIKV and DENV1 used only EDIII whereas those against the other DENVs used whole E proteins. The E proteins are larger and hence the amount of antigens on the coated plates would have fewer antigens than those coated with EDIII. The data from the ELISAs are not comparable. Showing this set of data in such a way is misleading.

11. There are several grammatical and typographical errors that need to be corrected.

Reviewer #2

(Remarks to the Author)
This manuscript described the development of a circular RNA vaccine to prevent ZIKV infection with reduced risk of causing ADE of subsequent DENV infection. The authors tested multiple circRNA constructs expressing Zika EDIII in monomer, dimer, or trimer conformation and determined the EDIII-Fc dimer is the most superior confirmation. The addition of NS1 into the vaccine formulation (EN) further improved maternal passive protection to neonatal mice from brain damage/inflammation than vaccine with EDIII-Fc or NS1 circRNA alone. The EN circRNA molecules encapsulated in LNP separately (EN(LNP)) or together (EN(RNA)) resulted in similar immunogenicity and protection outcomes. However, 2 doses of maternal immunization were required to provide complete protection from detectable ZIKV infection in neonatal mouse brains. Using an adult INFAR-/- mouse model, they also demonstrated both EN formulations elicited rapid protection (12 days) against illness and reduced viremic RNA post ZIKV infection.
To further optimize the EN(LNP) vaccine candidate, the PIE elements of the plasmid were modified. The optimization improved translation of EDIII, NS1, and Gluc in transfected cells and the optimized (opt) EN(LNP) provided better immunogenicity and protection than the prototype EN(LNP). A single high (20-ug) dose of the opt-EN(LNP) provided complete protection in both passive and active immunization mouse models. A longer protection duration was observed in neonatal mice born 11 weeks after maternal immunization with a high dose of the opt vaccine. Most of the study results supported the conclusion that a single dose of the opt-EN(LNP) ZIKV vaccine afforded high protection efficacy against ZIKV in the 2 tested mouse models. The manuscript detailed the steps from selecting an optimal EDIII conformation, rationale of including NS1, and optimization of the vaccine to be effective with a single dose. Overall, the study is very interesting and provides new knowledge in use of the circRNA platform for ZIKV vaccine development. However, there is a major concern related to the ADE evaluation of the vaccine and the strong conclusions regarding the vaccine ADE potential in the manuscript.
Based on the rationale of the viral antigens used for the vaccine, it is reasonable to speculate that the vaccine would induce less ADE-prone cross-reactive Abs to DENVs than other vaccine using prM-E or whole E. However, the in vitro and in vivo ADE evaluation conducted in the study had some significant flaws and limitations to really support their strong conclusion that the vaccine would be ADE-free. The title, abstract, and many areas of the manuscript all emphasized that the vaccine has proven to be: “without enhancing dengue severity”, “without DENV ADE”, and “ADE-free”, but I have questions regarding the ADE experimental data used to support such conclusion.
List of the specific concerns/comments:
1. The manuscript writing is fairly clear and understandable but seems somewhat choppy in some areas. Some minor editorial modification can be useful to make the manuscript more concise with smoother flow and connection, while providing more clarification in some areas to improve the reading experience.
2. In vitro ADE evaluation (Fig 4b and 6g-i): should the y-axis be expressed in percent (%)? Below are questions in the data presented in the graphs.
a. The “relative infectivity” normalized to “peak of ZIKV infection” in the Fig 4 and 6 did not make sense for presenting the DENV ADE curves. The relative infectivity should be calculated against infectivity of each virus without Ab treatment (base line infectivity for each specific virus in the graph). E.g., DENV-1 infectivity (without sera) would be the 100% for the DENV-1 graph. In our experience, ZIKV typically replicates much faster than DENVs in mammalian cells, so using the peak ADE ZIKV infectivity by diluted Zika sera to normalize the graph for all viruses would result in very low DENV infectivity curves as shown in 4b. Due to its slower replication kinetics than ZIKV, there would be less DENV-positive cells than the ZIKV-positive cells detected by flow cytometry from samples collected at the same time point (2 days pi). Also, Line 636 indicated the moi used for ZIKV and DENV-3 was 5-fold higher (at 1) than other DENVs (at 0.2). The higher moi used for ZIKV

infection would further skew the results for a much higher ZIKV infectivity than those DENVs infected at lower moi. The skewed graph scale made the interpretation of the DENV ADE curves difficult.
b. Even with the flawed ADE curve presentation, these graphs suggested that there were DENV ADE effects at the lowest dilutions of the vaccine sera. The full DENV ADE curves were not captured from the tested vaccination sera dilution range, and there is no standard Ab quantity (e.g. protein mass) to be used for X-axis. Therefore, it is difficult for an accurate ADE potential assessment or comparison from the graphs.
On a good note, Fig 4a demonstrated that the IgG elicited by vaccine candidates had significantly less proportion of DENV cross-reactive Ab than the ZIKV control (using anti-Zika IgG as the total IgG). Therefore, Fig 4a could support the notion that the vaccine candidates elicited significantly less DENV ADE-prone Abs than the live ZIKV infection. Unfortunately, similar anti-ZIKV and anti-DENV IgG were not measured for the final optimized EN(LNP). Would suggest to also conduct this analysis for the opt vaccine in Fig 6.
3. Fig 4 c-h: The in vivo ADE evaluation of the prototype vaccine candidates. The data was used to suggest passive transfer of vaccination sera or Ab elicited by vaccination did not enhance the severity of illness or enhance viremic RNA after DENV-2 challenge. In my opinion, the results could only suggest the single dose of the Ab in the 20ul sera (Fig.4c) or the Ab on 14 days after immunization of the vaccine (Fig 4f) did not cause ADE illness by DENV-2 challenge. Since the Ab concentration is an important factor for ADE, the single passive-transferred sera dose (4c) or the single DENV2 challenge time point (4f) may not be suitable for ADE evaluation of the vaccines. It might be better to use another non-replicative Zika vaccine with ADE-prone antigens (e.g. inactivated ZIKV, recombinant E, or VLP) as a positive ADE control for the ADE study.
The blood Ab titers to ZIKV and DENV2 at time (2wks) of challenge might enable better interpretation, but they were not measured in the study. However, based on the Fig 2g study, only 1 animal has measurable ZIKV nAb 2wks after 1st immunization of EDIII-Fc which suggests the 2wks time point for the DENV2 ADE evaluation might be too early (e.g. Ab levels were below the required threshold for ADE at the time of DENV2 infection). Interestingly, the in vivo ADE evaluation was not conducted for the final optimized vaccine candidate. Therefore, there is no animal ADE data to support the conclusion that the single-dose opt-EN(LNP) would not cause severe DENV illness as suggested in the manuscript title.
Mouse models are not very good for DENV or flavivirus ADE evaluation in general. Many flaviviruses without any evidence of causing ADE in humans would cause ADE in mouse models. Many biological components involved in ADE are very different between mouse and human, so caution should be exercised in using mice or in vitro ADE outcomes for human vaccine candidates. Discussions in the limitations of the ADE models should at least be included. Suggest removing or significantly toning down statements, such as “without DENV ADE”, “ADE-free”, and other similar claims throughout the manuscript.
4. The title: Suggest removing “without enhancement of dengue severity” and add “in mice” at the end or “prevents mice from” Zika virus infection.
5. Line 94: Need more background and citation of the IgG1 FC fusion to form EDIII dimer. Would there be any other immunogenic effect from the fusion Fc?
6. Line 104-110: please provide more background/introduction regarding the T4 ferritin for EDIII trimer formation, PIE method, and the reasons for the specific PIE element choice in the prototype plasmid Ana (e.g., why iCVB3 IRES was chosen, what are the spacer sequences used in this plasmid).
7. Line 118: Suggest adding a brief summary of the encapsulation efficiency and sizes of the circRNA-LNP, as they were mentioned in the Methods (line 483-486).
8. Lines 150 and 166: Suggest specifying what protective effects are being tested at the beginning of each section, such as protection of neonatal mice through passive maternal immunity (line 150) versus protection adult mice by active immunization (line 166).
9. Line 153, 553, and Fig 2a: Line 153 states mice were mated 3 weeks after immunization but indicated 2 weeks in Fig 2a and line 553.
10. Line 263-269 and 339-342: More background and discussion of the optimized PIE elements in the construct should be provided. What is the rationale to change to these element sequences (e.g., reason of change the IRES to iHV-B3 with Apt-eIF4G, change of spacer UTRs, etc.)?
11. Line 273: Why was the EN(RNA) vaccine taken out of contention against the EN(LNP) vaccine at this point? Both vaccines provided similar immunogenicity and protection, so it would be helpful to provide the reason of the EN(LNP) selection for further optimization development. Is the selection based on certain advantage (e.g., higher LNP encapsulation rate, easier vaccine formulation) or just by chance?
12. Line 562: should define FNT.
13. Line 588: change “teasing” to “passing”?
14. Line 865, Fig 2a: Line 865 states IFNAR -/- mice were challenged 2 weeks after 2nd immunization. Figure 2a says they were challenged 3 weeks after immunization.
15. Figure 3A: The mating schedule is a little unclear from this image. Suggest changing the branch out point immediately after 1st immunization (before 2wks). Based on where the two lines branch out at the current scheme, it looks like two rounds of mating occurred in the 2-dose study line.
16. Sup Fig 1a and 9a: what is the meaning of “optional” indicated in both plasmid figures?
17. Sup Fig 1h and 9e: It would be useful to provide a purpose of the junction site PCR and sequence validation.
18. nAb was measured after immunization, but was never measured after virus challenging from the survivors. The post challenge nAb titers would be useful to determine whether the vaccine provided a solid sterile immunity (no boost effect of the post challenge nAb titer). It is somewhat surprising that it was not done for the vaccine study.

Reviewer #3

(Remarks to the Author)
Nature manuscript # 492562_0

In the manuscript entitled " A single dose circular RNA vaccine completely prevents Zika virus infection without enhancing dengue severity" by Liu et al., the authors developed ZIKV vaccines in a form of circular RNA encoding the envelop protein domain III and NS1 and reported the efficacy of the vaccine candidates using C57BL/6 and IFNAR-/- mouse strains. The study results showed that maternal immunity elicited by circRNA encoding dimeric EDIII, fused to human IgG1 Fc fragment (EDIII-Fc) were superior to circRNA encoding monomeric or trimeric EDII in terms of activation of germinal center reactions, facilitating high neutralizing antibody production and induction of CD4/CD8 T cells producing cytokines. One-day-old neonates born from dams immunized with either one- or two dose vaccines survived after lethal challenge but do not prevent viral infection and morbidity. CircRNA encoding either ZIKV EDIII-Fc or NS1 alone provides partial protection against ZIKV infection and developmental retardation. Further, optimized circRNA EN(LNP), which increases stability and protein translation, reduced viral RNA burden in the neonatal brains and increase protective efficacy that was durable up to 12 weeks. The authors concluded that two-dose immunizations of circRNA EN (LNP) provide full protection against ZIKV infection while avoiding ADE of DENV, but not by a single dose immunization. The results highlight that circRNA could be a platform for a safe and effective flavivirus vaccine.

Major comments:

Overall, the manuscript is well-written. The studies were well- designed with proper controls. The authors made thorough efforts to improve the efficacy of LNP encapsulated optimized circRNA vaccine encoding EDIII and NS1. The study results showed promising outcomes of protection of adult and neonate C57BL/6 and IFANR-/- mice against ZIKV infection while avoiding possible ADE of DENV, which is a significant risk factor of flavivirus infection. However, I have a few concerns regarding data presentation, and the conclusion in the discussion was overstated and not fully supported by the data presented in the current version of manuscript. I suggest that the authors revise the conclusion to adhere to the results. See below the specific comment below.

Additional comments:

Figure 1f. Virus-binding antibody titers (d) were measured by one (ELISA) and neutralization antibody titers were measured by the other assay (FNT). Although the readouts of antibody response are reciprocal serum dilutions, it doesn't justify presenting the ratios of neutralizing antibodies to virus-binding antibodies (f). Virus-neutralizing antibodies are not exclusively IgG isotypes. IgA and IgM can be neutralizing yet virus-binding antibodies in this study measured IgG only not total antibodies. I doubt that the simple mathematical calculation of the ratio of serum titers is accurate and acceptable in the field. Without a thorough investigation using a method/platform to bridge two assays with well-established standard controls, the current form of data presentation can be misleading and inaccurate. If there are references supporting such data presentation, please add them or provide justifications. Otherwise, I suggest deleting Fig 1f. However, the authors may discuss the result and point out that EDIII-Fd induced robust anti-ZIKV antibody response but poorly neutralizing relative to antibody response induced by EDIII-Fc.

Lines 163-165: the authors stated that "…, including paralysis of limbs and tails, retardation of brain growth, meningeal thickening…." and presented brain mass and viral titers in tissues (Suppl. Fig3) but the evidence/data to support limb & tail paralysis and meningeal thickness are missing. According to histology in Figures 2f, 3f, and 4f, the immunized groups show less cellular infiltration in the brain tissue and possibly less necrosis than in the brain of the PBS group, in which dura membrane blebbing and cellular infiltration seem to be pronounced. It is difficult to know whether ZIKV induces changes in meningeal thickness in the representative image of each of the groups. No measurable data supporting meningeal thickening in the brain, nor the paralysis scores of limbs and tails were presented. Either provide the supporting data or revise.

Line 207-208/Fig 2d, " EDIII-Fc… prevented ZIKV-caused growth delay and mortality, alleviated neurological disorders,…." For clarification, indicate the score of 14 is the maximum severity of limb and tail paralyses per mouse either in figure legend or in the main body.

In Fig 2, add a line with "C57BL/6" above the panel of Fig2b-e and a line with "IFNAR-/-" above the panel of g-j.

Line 232, clarify what do the authors meant by "more than half showed"; does this mean more than half of the mice within the EDIII-Fc group?

In Fig 5h and other figures (2f, 3f and 3h), the presentation of compact images side-by-side is difficult to identify the point of the interest other than the increase of cellular infiltration possibly also cellular necrosis, no localization of viral proteins-particularly to readers with no pathological background. Add a short description of the point of interest indicated with boxes, arrows, or arrowheads that would be helpful to understand the presence or absence of pathology caused by viral infection and protection, respectively.

To make it easy to follow, add a line with the indication of " IFNAR-/-" below Fig 5h and above the panel of i-k in order to separate data from C57BL/6 mice and IFNAR-/- mice.

Line 304/ Fig6 e: At 15 dpi, 2 out of 13 pups (15%) after optEN(LNP) and 6 out of 7 pups (86%) after 20ug pEN(LNP) still showed substantially high levels (~10^8 copies) of viral RNA (Fig6e) with measurable neurological score (Fig 6d). Although the results presented "promising potential" of Opt EN(LNP) but did not support full nor complete protection against ZIKV infection at 13 weeks post-vaccination. The author shouldn't ignore these unprotected outliers and may discuss the result.

Line 307-8, the authors state that "Optimized circRNA vaccine has the potential to trigger…", which is likely, but data present the duration of protective immunity up to 12 weeks, which is relatively short. Consider revising "Optimized circRNA vaccine induced protective immunity lasting at least up to 12 weeks" to adhere the results presented.

Line 314, The statement "….the combination of EDIII-Fc and NS1 circRNAs confers complete protection in two mouse models (Fig. 3, 5)." is not correct. In Figure 3e, 3 out of 7 C57BL/6 mice (41%) after single dose immunization with EN(LNP) showed substantially high levels (108 copies/gram) of viral RNA. Only after two-dose immunizations with EN(LNP), mice were protected against infection but not after EN(RNA) immunization. In IFNAR-/-, viral RNA was also detected in all mice after single dose immunization (suppl. Fig 6g). I agree that the results presented are promising but, in the discussion, the conclusion remarks are overstated. Revise the conclusive statement adhering to the results presented in this manuscript.

Line 328, "our circRNA vaccine is quite promising …" sounds colloquial. I suggest revising "our circRNA vaccine produces promising results…"

Line 356, "..., but barely lymphocytes, can be vertically…" "barely" is not correct. In healthy conditions, maternal lymphocytes are not on the fetal side. Please revise.

Line 383-392, the authors described that the mouse adaptive DENV2 generation from the parental 16681 strain was obtained after 3 passages in the neonatal brain. This is relatively low passages compared with the mouse adaption achieved 10 passages of a parental strain, PL0468 in AG129 mice (DOI: 10.1128/JVI.00062-06). Relative to the old method, 3 passages in B6 neonates seem to be a huge improvement. Provide GenBank accession no. of the mouse-adaptive DENV. In addition to sequence analysis for identification of site mutations (supplementary Figure 8a), the authors should provide supplementary results showing productive DENV infection in immunocompetent or IFNAR-/- mice. If the characterization and pathogenesis of the current mouse-adapted DENV 2 were already published, please provide references. The citation 46 alone is insufficient.
Line 562, spell out "FNT" as it appears the first time in the main body of the manuscript.
Line 641, provide the source/vendor of DENV monoclonal antibody clone 8-4, I am not sure if the clone is correct.

Reviewer #4

(Remarks to the Author)
I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
The authors have addressed all the major concerns raised on their original manuscript. This revised version now reads well and provides a balanced interpretation of the data.

Reviewer #2

(Remarks to the Author)
We (primary reviewer and co-reviewer) thank the authors for the thorough responses including new experiment data and revision of the manuscript. All our concerns have been adequately addressed, and the revised manuscript has been greatly improved in clarity and provides significant findings in developing and optimizing a circular RNA ZIKV vaccine. Below are 2 minor comments of the revised manuscript:
1. Fig 8a and b: Missing the ZIKV sera data in Fig 8a or b. Since the ZIKV sera was used in Fig 8c as the in vitro ADE positive control, it would be better to include its overall IgG titers to various viral E antigens in Fig 8a and/or 8b (as shown in Fig 5a).
2. Figure legends for all brain H&E staining images: Shouldn't black arrows indicate cortex laminar necrosis, and cyan blue arrows lymphocyte infiltration? All legends for these figures indicated opposite.

Reviewer #3

(Remarks to the Author)
The authors have addressed the reviewers' comments made on the previous submission. Additionally, the authors have made an extensive revision that improved clarity and provided additional supporting information. The results presented in this manuscript strongly support the conclusions.

I have a few minor suggestions and corrections for citations as indicated below:

line 159, " .. plasma cells were significantly detected…" sounds awkward and I suggest revising 'significantly high' or '..significantly higher… than in the mice immunized with EDIII-Fd or the others'.

line 167, instead of "but", use either 'and' or 'while these were…'

lines 297 and 310, in the description of passive ZIKV-immune sera transfer, there is a discrepancy in the volume of serum samples, "20 uL" on line 297 ' 200 uL of diluted serum" on line 310 and Figure legend 5C. Please be consistent.

line 449, 'favorable' may be better than "promising"

line 547, Vero cells is CCL-81 not "NCCL-81"

line 562, "1x10^4 FFUs of DENV2", delete 's' in FFUs

line 813, change from "qRT-qPCR" to "qRT-PCR" consistent with line 812

Comments on citations:

line 52, Amaya et al., 2023 is inappropriate. Please correct.

line 515, the statement "…. the pregnancy mouse models used in other studies 61, 62" including citation 62 is incorrect. Reference 62 used a litter-bearing non-human primate, marmoset pregnancy model not a mouse model. Authors may replace reference 62 with another immunocompetent mouse pregnancy model, human Stat2 knock-in (hSTAT2KI) mouse (PMID: 38360793). hSTAT2KI mouse model recapitulates vertical transmission of ZIKV during pregnancy while ZIKV causes fetal demise in pregnant C57BL/6 mouse model (reference 61). Alternatively, authors may revise the statement from "…the pregnancy mouse models…" to "the preclinical pregnancy models" referring mouse and marmoset models.

Reviewer #4

(Remarks to the Author)
I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

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Reviewer #1 (Remarks to the Author):

This manuscript by Liu and colleagues reports a preclinical study into a circular RNA vaccine to prevent Zika virus (ZIKV) infection. The authors compared 3 different constructs of circRNA to identify one that fused EDIII of ZIKV with a human IgG1 Fc fragment to form an EDIII dimer. They show that circRNA expressing EDIII-Fc produced superior immunogenicity compared to the other 2 constructs in adult female mice, and this superior immunogenicity translated to better protection against lethal ZIKV infection in pups from the vaccinated females. They also showed that the antibody response produced from vaccination cross-reacted minimally with DENVs and did not enhance DENV infection to levels seen with antisera against ZIKV. Finally, they showed that with additional modification, such as the use of IRES of iHRV-B3, the vaccine had the potential of being a single dose application with even better protection compared to their original construct. They concluded that their findings set a foundation for developing circRNA flavivirus vaccines.

This is an interesting study although there are several concerns that need to be addressed. These are:

1. Congenital Zika syndrome (CZS) is caused by ZIKV infection in expectant mothers, where transplacental transmission to the developing fetus dysregulates normal CNS development. In this study, ZIKV is inoculated to pups from vaccinated dams. Although I do not think the study design makes the findings less robust, a discussion on the differences in infection route between the experimental approach and real-world infection, along with their implication on vaccine-mediated protection against CZS should be clearly discussed.

Response: We agree with the notion that ZIKV infection in pregnant animals, such as immunocompetent mice and marmosets, are more relevant to the CZS observed in real world (Kim IJ et al., NPJ Vaccines 2022; Kim IJ et al., NPJ Vaccines 2024). Though our study's model of maternal immunization and neonatal challenge doesn't mimic human CZS as closely as pregnancy models in other studies (Kim IJ et al., NPJ Vaccines 2022; Kim IJ et al., NPJ Vaccines 2024), it may represent a stringent model for evaluating the protective efficacy of maternal immunity. Unlike the pregnant mice

models wherein both nAbs and T cell responses contribute to restricting the inoculated ZIKV from crossing the placental barriers (Kim IJ et al., NPJ Vaccines 2024; Ngono AE et al., Sci Adv 2020), in the challenged pups only the nAbs inherited from the dams can confer protection. In addition, it has been reported that in pregnant mice, the monocytes or macrophages in the placenta play significant roles in inhibiting ZIKV transmission to the fetus, another barrier absent in the challenged pups (Winkler CW et al., J Immunol 2020). In this respect, our results support that EN(LNP), especially the optimized version, gives rise to protective immunity of high magnitude and quality, thereby conferring complete protection against ZIKV infection in this stringent model (Revised Fig. 6 and Fig. 7). We have added this discussion to the revised manuscript (lines 514-526), following your insightful suggestion.

2. Although the authors argued for the merits of circRNA, the dose used in this study – 20 microgram – is high. Conventional COVID-19 nucleoside-modified linear mRNA vaccines were showed efficacy against lethal SARS-CoV-2 infection in mice at a dose of 5 micrograms. The higher dose would mean not only higher cost, it could also lead to more adverse events and increase the risk of severe adverse events. Moreover, there is no comparison between circRNA and linear, nucleoside modified mRNA that express the same 3 different EDIII constructs. Thus, while the authors argued for the merits of circRNA, there is no data to show that circRNA is a better system to deliver the EDIII-Fc transgene.

Response: Our initial purpose is to determine whether the combination of EDIII-Fc and NS1 encoded by circRNA, an emerging RNA technology with enhanced stability over linear mRNA (Ren L et al., Commun Biol 2022), can confer protection against ZIKV infection. Starting from this, we initially focused on the design and evaluation of circRNA vaccines but not their linear mRNA counterparts. Nevertheless, we have performed a direct comparison between the optimized circRNAs and their linear mRNA counterparts in both the translational level in cell cultures and the immunogenicity in immunized mice. Optimized circRNAs encoding either EDIII-Fc or NS1 appeared to have lower translation levels compared to their respective m1ψ-modified mRNA

counterparts (Revised Supplementary Fig. 10). At 2 weeks after a single-dose immunization, optimized circRNAs elicited lower titers of anti-E IgG (1.8- to 2.4-fold), anti-NS1 IgG (3.4- to 3.9-fold), and anti-ZIKV nAbs (2.0- to 3.5-fold) compared to the same dose of each respective mRNAs (Supplementary Fig. 10). These results, along with those shown in revised Fig. 6 and Fig. 7, suggest that the optimization effectively enhances the antigen production and immunogenicity of circRNAs, albeit still being weaker than that of m1ψ-modified mRNAs at the tested doses and timeframe. We have added these results in the revised manuscript (lines 331-342, revised Supplementary Fig. 10), and discussed the drawbacks that need to be resolved as well as the modifications that may further improve the immunogenicity and protective efficacy of current circRNA vaccines (lines 469-477 in the revised manuscript).

3. Lines 56-57. This statement is not true. There are Zika vaccine candidates that have completed phase I clinical trial with useful safety outcomes. Their development has been stalled by the global low incidence of Zika cases. Please correct this statement.

Response: We have corrected this statement as "Currently, there are still no ZIKV vaccines approved for clinical use" (line 58 in the revised manuscript), following your valuable suggestion.

4. Lines 58-59. Zika vaccine development has not been impeded by the concern of ADE of DENV infection. Please see explanation in comment #3.

Response: We have revised this sentence as "An important safety concern for vaccine development against ZIKV is the antibody-dependent enhancement (ADE) of infection between ZIKV and DENV" (lines 59-60 in the revised manuscript), following your valuable suggestion.

5. Line 74. It is not helpful to mix the pathology of dengue with that of Zika. Coagulopathy and vascular leakage is more pertinent to dengue than Zika.

Response: We have described the pathogenic effects of ZIKV NS1 more specifically, as the following: "Secreted NS1 increases the permeability of umbilical vein and brain

endothelial cells, and thus causes vascular leakage in the placentas and brains" (lines 78-79 in the revised manuscript). We also provided new references to support this description (Puerta-Guardo H et al., Cell Rep 2019).

6. Line 122. Description of results in this line and elsewhere should be described as objectively as possible. Please avoid superlatives, such as "greatly".

Response: Following your kind suggestion, we have more objectively described this result, as the following: "IgG titers increased by 3.7-fold in EDIII-Fc-immunized mice, by 1.7-fold in EDIII-Fd-immunized mice" (lines 145-148 in the revised manuscript). We have revised all the similar descriptions throughout the manuscript.

7. Lines 144-147 and Figure 1K. Although the differences in $CD8^+$ T cell response to the different vaccine constructs raises interesting questions on the expression or processing of the different EDIII constructs for MHC class I presentation, it raises concern for EDIII-Fc as a human vaccine. Human $CD8^+$ T cell epitopes are found mostly elsewhere in the ZIKV and flaviviral proteome. If the protection seen in subsequent experiments, particularly those on adult $IFNAR^{-/-}$ mice were to be mediated, even in parts, by $CD8^+$ T cells, then the vaccine efficacy seen in this mouse model may over-estimate the efficacy that can be expected in humans. Indeed, this possibility is raised by the data in Figures 4h and 5k, where the neutralizing antibody titers produced after vaccination were not at levels sufficient for sterilizing immunity, as evidenced by detectable viremia even for the most optimal construct. Control of infection at later time points could thus have been mediated by cellular immunity, given that we now know anamnestic T cell response is faster than B cell response (Painter et al, Nat Immunol 2023; 24:1711-1724).

Response: We agree with your opinion that $CD8^+$ T cell response contributes to the protective effects observed in the adult $IFNAR^{-/-}$ mouse model. However, the over estimation of the vaccine efficacy in humans might be minimal. In the adult $IFNAR^{-/-}$ mouse model, the protective immunity elicited by EN(LNP) or the optimized version are primarily comprised of four parts: EDIII-specific neutralizing antibodies, NS1-

specific protective antibodies (Wessel AW, et al., Nat Commun, 2020; Yu L et al., mBio, 2021; Bailey MJ, et al., Nat Commun, 2018), EDIII-specific CD8+ T cells (revised Fig. 1j), and NS1-specific CD8+ T cells (Grubor-Bauk B, et al., Sci Adv, 2019; Brault AC, et al., Sci Rep, 2017). Actually, the EDIII and NS1 used in this study contains several human HLA class I restricted epitopes, e.g., peptides "GPCKVPAQMAV" (named FSS-$E_{337–347}$), "ELDPPFGDSY" (named FSS/MR-$E_{377–386}$), "GPQRLPVPV" (named FSS/MR-$NS1_{99–107}$), and "DVEAWRDRY" (named FSS/MR-$NS1_{23–31}$). $CD8^+$ T cells specific for FSS-$E_{337–347}$ and FSS/MR-$NS1_{99–107}$, restricted by human HLA-B*0702, could be detected in HLA-B*0702 transgenic mice infected by ZIKV. Similarly, $CD8^+$ T cells specific for FSS/MR-$E_{377–386}$ and FSS/MR-$NS1_{23–31}$, restricted by human HLA-A*0101, could be detected in HLA-A*0101 transgenic mice infected by ZIKV (Wen J, et al., Nat Microbiol, 2017). These results imply that EN(LNP), or the optimized version, has the potential to trigger EDIII-specific and NS1-specific $CD8^+$ T cell responses in humans, at least in individuals carrying the HLA-A*0101 or HLA-B*0702 allele. We noted that the strength of the $CD8^+$ T cell response elicited by EDIII and NS1 in human might be distinct to that in mice, but the protective antibody responses could partially bridge the gap. The results in neonatal challenge model indicate that antibody responses are sufficient to suppress ZIKV infection in the brains and the occurrence of ZIKV diseases (Revised Fig. 4 and Revised Fig. 6). We have discussed this point in the revised manuscript (lines 527-543).

8. Line 248 and 384-392. Please specify the strain of mouse-adapted DENV-2 used in this study. What was the parental strain of this mouse-adapted DENV-2? Moreover, it is puzzling that the authors generated this strain by inoculating C57Bl/6 mice subcutaneously and could then harvest the virus from the brains of these animals. The reference they cited (Orozco et al, J Gen Virol 2012) used AG129 mice and passaged the virus for 11 cycles and not just 3 passages. Is this correct?

Response: In this study, we generated mouse-adapted DENV2 from DENV2 strain 16681 (GenBank No. U87411), which was derived from a dengue haemorrhagic fever patient. The passage were performed in 1-day-old C57BL/6 mice and Vero cells for 3

cycles. We have submitted the genomic sequences of the mouse-adapted DENV-2 in GeneBank (No. PQ008452).

It has been reported that a subcutaneous inoculation with DENV2 (NGC strain) led to lethal brain infection in 1-day-old C57BL/6 mice (https://www.fda.gov/media/168791/download). Another study showed that an intranasal inoculation with DENV2 also led to severe brain infection in 3-day-old suckling BALB/c mice, and live DENV2 virions could be isolated from the brain tissues and passaged on Vero cells (Qiu M et al., Viruses, 2022). The rapid occurrence of the mouse-adapted variant in our model might be attributed to the following reasons: i. The genome of DENV2 appeared to be less stable in Vero cells than in C6/36 cells (Chen WJ, et al., Intervirology, 2003), leading to a variety of quasi-species populations that can be selected in the infected mice; ii. The higher culture temperature of Vero cells (37 ℃) relative to C6/36 cells (usually 28 ℃) maybe also contribute to the occurrence of quasi-species population harboring virulence-imparting mutations (Modak A, et al., FASEB J, 2023); iii. Establishment of infection by DENV2 in neonatal brains need to break through more barriers compared to the method used by Orozco S et al (intravascular inoculation followed by viral isolation from the sera) (Orozco S et al., J Gen Virol, 2012; Shresta S et al., J Virol, 2006), thereby exerting stronger selective pressure.

We identified three amino acid substitutions in the mouse-adapted variant compared to the parental DENV2: NS1 K174N, NS2A L181V, and NS5 A196T. Although the exact contribution of each mutation to the virulence remains to be determined, an intracerebral injection of the adapted variant led to more rapid mortality in 1-day-old ICR mice compared the parental DENV-2 (Revised Supplementary Fig. 8b, c). We have added this result to the revised manuscript (lines 292-298).

9. Lines 250-260. These statements misrepresent the findings in Figures 4d, 4e and 4g. Even sera from vaccinated animals led mortality and viremia in this model. Moreover, p=0.9179 in the survival curve for EN(LNP) vaccinated mice vs uninfected controls is simply unbelievable given that over 50% of the mice died from infection.

Response: Actually, original Fig. 4d, 4e, 4g and 4h (revised Fig. 5d, 5e, 5g, and 5h)

showed that ZIKV sera, but not circRNA immune sera or active circRNA immunization, could enhance the infection and diseases of DENV2. We have revised Fig. 4c and 4f (corresponding to revised Fig. 5c and 5f) to more clearly describe the experiment design. Please note that the challenge assays herein were performed using mouse-adapted DENV2. Given that immunization with EN(LNP) or EDIII-Fc elicited only residual cross-reactive antibodies against DENV2 (revised Fig. 5a), it is reasonable that either the immune sera or active circRNA immunization showed no significant enhancement of or protection against DENV2 infection. We have revised the related statements to present the findings more clearly (lines 300-306 in the revised manuscript).

In the original Fig. 4d and 4g (corresponding to revised Fig. 5d and 5g), the comparisons were conducted between the PBS group (revised Fig. 5d) or the LNP group (revised Fig. 5g) and the rest groups. Thus, the "p = 0.9179" here indicates that there was no difference in the survival rates between EN(LNP)-immunized mice and LNP-immunized mice after DENV2 challenge. We have these figures and the related legends to present the statistical analysis more clearly (line 1216 in the revised manuscript).

10. Figure 4a and Lines 914-916. This set of data cannot be shown in such a way. The IgG titers were measured using different approaches. Those for ZIKV and DENV1 used only EDIII whereas those against the other DENVs used whole E proteins. The E proteins are larger and hence the amount of antigens on the coated plates would have fewer antigens than those coated with EDIII. The data from the ELISAs are not comparable. Showing this set of data in such a way is misleading.

Response: We have noted that the original Fig. 4a is somewhat misleading. We have revised this figure by showing the IgG titers against different antigens separately. Please refer to the revised Fig. 5a.

11. There are several grammatical and typographical errors that need to be corrected.

Response: We have tried our best to correct the grammatical and typographical errors in the revised manuscript.

Reviewer #2 (Remarks to the Author):

This manuscript described the development of a circular RNA vaccine to prevent ZIKV infection with reduced risk of causing ADE of subsequent DENV infection. The authors tested multiple circRNA constructs expressing Zika EDIII in monomer, dimer, or trimer conformation and determined the EDIII-Fc dimer is the most superior confirmation. The addition of NS1 into the vaccine formulation (EN) further improved maternal passive protection to neonatal mice from brain damage/inflammation than vaccine with EDIII-Fc or NS1 circRNA alone. The EN circRNA molecules encapsulated in LNP separately (EN(LNP)) or together (EN(RNA)) resulted in similar immunogenicity and protection outcomes. However, 2 doses of maternal immunization were required to provide complete protection from detectable ZIKV infection in neonatal mouse brains. Using an adult INFAR-/- mouse model, they also demonstrated both EN formulations elicited rapid protection (12 days) against illness and reduced viremic RNA post ZIKV infection.

To further optimize the EN(LNP) vaccine candidate, the PIE elements of the plasmid were modified. The optimization improved translation of EDIII, NS1, and Gluc in transfected cells and the optimized (opt) EN(LNP) provided better immunogenicity and protection than the prototype EN(LNP). A single high (20-ug) dose of the opt-EN(LNP) provided complete protection in both passive and active immunization mouse models. A longer protection duration was observed in neonatal mice born 11 weeks after maternal immunization with a high dose of the opt vaccine. Most of the study results supported the conclusion that a single dose of the opt-EN(LNP) ZIKV vaccine afforded high protection efficacy against ZIKV in the 2 tested mouse models. The manuscript detailed the steps from selecting an optimal EDIII conformation, rationale of including NS1, and optimization of the vaccine to be effective with a single dose. Overall, the study is very interesting and provides new knowledge in use of the circRNA platform for ZIKV vaccine development. However, there is a major concern related to the ADE evaluation of the vaccine and the strong conclusions regarding the vaccine ADE potential in the manuscript.

Based on the rationale of the viral antigens used for the vaccine, it is reasonable to

speculate that the vaccine would induce less ADE-prone cross-reactive Abs to DENVs than other vaccine using prM-E or whole E. However, the *in vitro* and *in vivo* ADE evaluation conducted in the study had some significant flaws and limitations to really support their strong conclusion that the vaccine would be ADE-free. The title, abstract, and many areas of the manuscript all emphasized that the vaccine has proven to be: "without enhancing dengue severity", "without DENV ADE", and "ADE-free", but I have questions regarding the ADE experimental data used to support such conclusion.

Response: We appreciate your insightful comments and suggestions. We have noted that there are several over statements in the original manuscript. We thus performed additional experiments, mainly regarding the ADE risks associated with the optimized EN(LNP), to provide more evidence supporting the reduced ADE risks of the vaccine design, please refer to the revised Fig. 8. The immune sera of optimized EN(LNP) collected at either 2 or 8 weeks after immunization contain only residual DENV-reactive antibodies (revised Fig. 8a, b). Similar to the observations regarding the prototype EN(LNP), neither the immune sera of optimized EN(LNP) nor the active immunization showed any signs of ADE effects on DENV2 infection (revised Fig. 8c-h). Nevertheless, considering the difference between the pathogenesis of ADE in humans and that in mice models, we toned down the conclusion by specifying the settings of the detection. Please see our point-by-point responses to your comments, as listed below.

List of the specific concerns/comments:

1. The manuscript writing is fairly clear and understandable but seems somewhat choppy in some areas. Some minor editorial modification can be useful to make the manuscript more concise with smoother flow and connection, while providing more clarification in some areas to improve the reading experience.

Response: We are grateful for your kind suggestions. We have carefully revised the manuscript to improve the clarity, coherence, and flow.

2. In vitro ADE evaluation (Fig 4b and 6g-i): should the y-axis be expressed in percent

(%)? Below are questions in the data presented in the graphs.

a. The "relative infectivity" normalized to "peak of ZIKV infection" in the Fig 4 and 6 did not make sense for presenting the DENV ADE curves. The relative infectivity should be calculated against infectivity of each virus without Ab treatment (base line infectivity for each specific virus in the graph). E.g., DENV-1 infectivity (without sera) would be the 100% for the DENV-1 graph. In our experience, ZIKV typically replicates much faster than DENVs in mammalian cells, so using the peak ADE ZIKV infectivity by diluted Zika sera to normalize the graph for all viruses would result in very low DENV infectivity curves as shown in 4b. Due to its slower replication kinetics than ZIKV, there would be less DENV-positive cells than the ZIKV-positive cells detected by flow cytometry from samples collected at the same time point (2 days pi). Also, Line 636 indicated the moi used for ZIKV and DENV-3 was 5-fold higher (at 1) than other DENVs (at 0.2). The higher moi used for ZIKV infection would further skew the results for a much higher ZIKV infectivity than those DENVs infected at lower moi. The skewed graph scale made the interpretation of the DENV ADE curves difficult.

Response: We have noted that the normalization method used in the original Fig. 4 and Fig. 6 is inappropriate. In the revised Fig. 5b and Fig. 8c, we present the frequencies of infected cells at each sera dilution alternatively, as did by multiple previous studies (Goncalvez AP et al., Proc Natl Acad Sci U S A, 2007; Durham ND et al., eLife, 2019; Garg H et al., BMC Immunol, 2021).

b. Even with the flawed ADE curve presentation, these graphs suggested that there were DENV ADE effects at the lowest dilutions of the vaccine sera. The full DENV ADE curves were not captured from the tested vaccination sera dilution range, and there is no standard Ab quantity (e.g. protein mass) to be used for X-axis. Therefore, it is difficult for an accurate ADE potential assessment or comparison from the graphs.

On a good note, Fig 4a demonstrated that the IgG elicited by vaccine candidates had significantly less proportion of DENV cross-reactive Ab than the ZIKV control (using anti-Zika IgG as the total IgG). Therefore, Fig 4a could support the notion that the vaccine candidates elicited significantly less DENV ADE-prone Abs than the live ZIKV

infection. Unfortunately, similar anti-ZIKV and anti-DENV IgG were not measured for the final optimized EN(LNP). Would suggest to also conduct this analysis for the opt vaccine in Fig 6.

Response: We agree with the notion that quantifying antibody (Ab) concentration is important for an accurate assessment of the ADE risks. However, directly comparing the concentrations of E-specific IgG in ZIKV sera and immune sera presents challenges. This is because ZIKV infection induces a distinct IgG profile compared to those elicited by EDIII immunization. The variety of IgG clones may differ in epitope recognition and binding activities. Hence, we utilized sera dilutions as a surrogate to compare the ADE effects between different sera, akin to the methods employed in several previous studies (Alwis R et al., Plos Pathog, 2014; Dai L et al., Nat Immunol, 2021; Yen LC et al., J Biomed Sci, 2023).

Following your valuable suggestion, we have tested the titers of DENV-reactive IgG in the optimized EN(LNP) immune sera collected at 2 or 8 weeks after immunization. The results have been illustrated in the revised Fig. 8a and 8b. Because ZIKV sera showed robust cross-reactivity at this timeframe (revised Fig. 5a and Supplementary Fig. 7), these results suggest that similar to prototype EN(LNP), optimized EN(LNP) also elicits significantly lower titers of DENV-reactive antibodies compared to ZIKV infection. These outcomes, combined with the revised Fig. 8d-f that demonstrated the absence of DENV2 ADE associated with optimized EN(LNP) sera, support that optimized EN(LNP) elicits only residual ADE-prone antibodies, which are insufficient to enhance dengue severity in mice models. We have added these results and a brief discussion in the revised manuscript (lines 404-419 and 495-503, respectively).

3. Fig 4 c-h: The *in vivo* ADE evaluation of the prototype vaccine candidates. The data was used to suggest passive transfer of vaccination sera or Ab elicited by vaccination did not enhance the severity of illness or enhance viremic RNA after DENV-2 challenge. In my opinion, the results could only suggest the single dose of the Ab in the 20ul sera (Fig.4c) or the Ab on 14 days after immunization of the vaccine (Fig 4f) did not cause ADE illness by DENV-2 challenge. Since the Ab concentration is an important factor

for ADE, the single passive-transferred sera dose (4c) or the single DENV2 challenge time point (4f) may not be suitable for ADE evaluation of the vaccines. It might be better to use another non-replicative Zika vaccine with ADE-prone antigens (e.g. inactivated ZIKV, recombinant E, or VLP) as a positive ADE control for the ADE study. The blood Ab titers to ZIKV and DENV2 at time (2wks) of challenge might enable better interpretation, but they were not measured in the study. However, based on the Fig 2g study, only 1 animal has measurable ZIKV nAb 2wks after 1st immunization of EDIII-Fc which suggests the 2wks time point for the DENV2 ADE evaluation might be too early (e.g. Ab levels were below the required threshold for ADE at the time of DENV2 infection). Interestingly, the *in vivo* ADE evaluation was not conducted for the final optimized vaccine candidate. Therefore, there is no animal ADE data to support the conclusion that the single-dose opt-EN(LNP) would not cause severe DENV illness as suggested in the manuscript title. Mouse models are not very good for DENV or flavivirus ADE evaluation in general. Many flaviviruses without any evidence of causing ADE in humans would cause ADE in mouse models. Many biological components involved in ADE are very different between mouse and human, so caution should be exercised in using mice or in vitro ADE outcomes for human vaccine candidates. Discussions in the limitations of the ADE models should at least be included. Suggest removing or significantly toning down statements, such as "without DENV ADE", "ADE-free", and other similar claims throughout the manuscript.

Response: We concur with your insights that the results depicted in the original Fig. 4c-h demonstrate the absence of DENV ADE under the particular conditions tested. However, these findings, along with the additional data presented in revised Fig. 8d-i, demonstrated the significantly reduced ADE risks associated with circRNA vaccines. In the passive transfer study, our utilization of 20 µl of immune sera aimed to replicate the *in vitro* conditions presented in the revised Fig. 5b and Fig. 8c. Given that an adult mouse typically has 1.5-2 ml of blood, the infusion of 20 µl of immune sera results in an approximate dilution at 1:100. At this dilution, the immune sera seemed to promote DENV infection in K562 cells, albeit to a lesser degree than ZIKV sera. Given that the transfer of such a volume of immune sera does not cause DENV2 ADE in mice, the

residual cross-reactive antibodies elicited by EDIII-Fc or EN(LNP), at least at this timeframe, are insufficient to induce ADE *in vivo*. Following your insightful suggestion, we tested the ADE risks of the immune sera of optimized EN(LNP), which were collected at 2 or 8 weeks after immunization and contained high titers of anti-ZIKV nAbs (revised Fig. 6d and Fig. 7b), as shown in the revised Fig. 8a-f. These sera, similar to the immune sera of prototype EN(LNP), contained residual cross-reactive antibodies and demonstrated ADE effects on DENV infection in K562 cells at dilutions below 1:2500. However, the infusion of 20 μl of these sera still did not cause obvious DENV2 ADE in mice. These findings further support that the antibody responses elicited by the circRNA vaccine pose minimal, if any, ADE risks in mice.

Regarding the active immunization experiment, we have noted that the evaluation at a single time point would give limited information, although E-specific IgG antibodies have become detectable at 2 weeks after immunization (revised Supplementary Fig. 3c and Supplementary Fig. 11f). We thus examined the ADE risks of active immunization with optimized EN(LNP) at 3 weeks after immunization. Herein, we used mice immunized twice with inactivated ZIKV as controls (treated with 0.2% β-propiolactone, combined with Aluminum adjuvant, and administered at 1×10^5 FFUs per mouse), according to your kind suggestions. Following the challenge with mouse-adapted DENV2, we observed no signs of DENV2 ADE in mice immunized with optimized EN(LNP) (revised Fig. 8g-i). In contrast, mice immunized with inactivated ZIKV exhibited more severe symptoms and increased viremia, compared to those receiving empty LNPs (revised Fig. 8g-i). These findings underscore that, under the conditions tested, immunization with optimized EN(LNP) unlikely induces DENV2 ADE in mice.

Although immunocompromised mice models have been employed to explore the ADE risks of antibodies and immune sera (Reviewed in Yang X et al., Viruses 2022), current mouse models, including those with a humanized immune system, fail to fully replicate the hallmark features of severe dengue in humans. There are significant differences between human and mouse immune responses, such as antibody response profiles, expression and distribution of FcγRs, as well as inflammatory pathways. Consequently, we acknowledge that a comprehensive evaluation of the ADE risks of

circRNA vaccines necessitates an appropriate animal model that closely mirrors human antibody responses and DENV pathogenesis. Such evaluation should be conducted across multiple dosages over an extended period to ensure cross-reactive antibody titers reach the threshold necessary for the occurrence of ADE. Accordingly, we have toned down certain claims and specified the assay conditions in the revised manuscript. We have also included a discussion section to outline the limitations of the ADE models used in this study (lines 506-513).

4. The title: Suggest removing "without enhancement of dengue severity" and add "in mice" at the end or "prevents mice from" Zika virus infection.

Response: Thank you for your valuable suggestion regarding the manuscript title. We agree that the addition of "in mice" helps to accurately reflect the scope of the study. Given that we have provided additional data to support the absence of DENV2 ADE in mice in the context of either passive transfer of optimized EN(LNP) sera or active immunization with optimized EN(LNP) (revised Fig. 8, see also our response to your comment 3), we revised the title as "A single-dose circular RNA vaccine prevents Zika virus infection without enhancing dengue severity in mice". We believe this title has summarized the main findings of this study clearly and precisely.

5. Line 94: Need more background and citation of the IgG1 FC fusion to form EDIII dimer. Would there be any other immunogenic effect from the fusion Fc?

Response: In addition to facilitate the dimerization of EDIII, which may enhance its avidity for B cell receptors (Hinke DM et al., Cell Rep 2022), the IgG1 FC fusion may extend the half-life of EDIII through neonatal Fc receptor (FcRn) recycling mechanisms (Suzuki T et al., J Immunol 2010; Kim MY et al., Trend Biotechnol 2024). Furthermore, this fusion also potentially improves the uptake and presentation of EDIII by FcγRs-bearing antigen presenting cells (APCs) in draining lymph nodes (Junker F et al., Front Immunol 2020; Kim MY et al., Trend Biotechnol 2024), thereby augmenting the immunogenicity. Following your insightful suggestion, we have incorporated this explanation, as well as the related references to the revised manuscript (lines 99-103).

6. Line 104-110: please provide more background/introduction regarding the T4 ferritin for EDIII trimer formation, PIE method, and the reasons for the specific PIE element choice in the prototype plasmid Ana (e.g., why iCVB3 IRES was chosen, what are the spacer sequences used in this plasmid).

Response: We have provided more details regarding the foldon domain and the PIE method in the revised manuscript (lines 116-121), in response to your valuable suggestion. Initially, our choice of the PIE method based on the group I catalytic intron from *Anabaena* pre-tRNA was influenced by its prominent effectiveness (over 95% circularization rate), as detailed in the study by Wesselhoeft RA et al (Nat Commun 2018). Their research highlighted that introducing permissive spacers between the 3' intron-exon juncture and the IRES, as well as between the coding sequence and the exon-5' intron junction, was critical to preserving the catalytic secondary structure of the intron. Furthermore, we opted for the iCVB3 IRES over the commonly used EMCV IRES based on its superior efficacy in facilitating cap-independent translation within circRNAs (Wesselhoeft RA et al., Nat Commun 2018). We added a very brief description about the IRES and spacers in the main text (lines 124-126). The sequences of the spacers have been added in the Methods section (lines 591-593).

7. Line 118: Suggest adding a brief summary of the encapsulation efficiency and sizes of the circRNA-LNP, as they were mentioned in the Methods (line 483-486).

Response: We have provided the encapsulation efficiency and sizes of the circRNA-LNPs used in this study, please see the revised manuscript (lines 141-143) and Supplementary Table 1.

8. Lines 150 and 166: Suggest specifying what protective effects are being tested at the beginning of each section, such as protection of neonatal mice through passive maternal immunity (line 150) versus protection adult mice by active immunization (line 166).

Response: We have added the specific purpose of the challenge assays at the beginning of each section in the revised manuscript (lines 177-178 and 196-197), akin to your kind suggestion. We also added a brief description of the purpose or specific

conditions at the beginning of several other sections to improve the coherence.

9. Line 153, 553, and Fig 2a: Line 153 states mice were mated 3 weeks after immunization but indicated 2 weeks in Fig 2a and line 553.

Response: Herein the mice were mated at 2 weeks after the final immunization. We have re-drawn the schematic diagram for each animal experiments in the revised Figures, to avoid any misleading information. We have also corrected the discrepancies in the revised manuscript.

10. Line 263-269 and 339-342: More background and discussion of the optimized PIE elements in the construct should be provided. What is the rationale to change to these element sequences (e.g., reason of change the IRES to iHV-B3 with Apt-elF4G, change of spacer UTRs, etc.)?

Response: During the progress of this study, Chen R et al reported a refined design of circRNAs which exhibits high translational activity (Nat Biotech 2022). In their research, they illustrated that the replacing of iCVB3 with iHRV-B3, incorporating a RNA-binding motif for poly(A)-binding proteins (PABPs) in the 5' upstream of iHRV-B3, inserting an eIF4G-recruiting aptamer (Apt-eIF4G) into the IRES, and adding the 3′ UTR of human α-globin 1 (HBA1) could dramatically enhance the translation of a variety of proteins by circRNAs. iHRV-B3 showed superior potency than iCVB3, and the inclusion of Apt-eIF4G, along with the 5' and 3' UTRs, may assist in recruiting eukaryotic initiation factors (eIFs) and other RNA-binding proteins, thus improving the initiation and strength of translation (Chen R et al., Nat Biotech 2022; Jeeva S et al., Vaccines (Basel) 2021). These studies informed our approach to optimizing circRNA vaccines. In the revised manuscript, we have expanded the background information on these optimization strategies (lines 319-326), and discussed the effectiveness of these enhanced elements further (lines 463-468).

11. Line 273: Why was the EN(RNA) vaccine taken out of contention against the EN(LNP) vaccine at this point? Both vaccines provided similar immunogenicity and

protection, so it would be helpful to provide the reason of the EN(LNP) selection for further optimization development. Is the selection based on certain advantage (e.g., higher LNP encapsulation rate, easier vaccine formulation) or just by chance?

Response: We chose EN(LNP) for further evaluation because of its simpler formulation process. Unlike EN(RNA), which requires multiple encapsulation steps for each specific ratio, EN(LNP) allows for flexible adjustments between the ratios of EDIII-Fc circRNA and NS1 circRNA with just two encapsulation steps. Although in this study we evaluated only one ratio, 1:1, in future studies we plan to test different ratios to achieve a further optimization. This is especially important for a multivalent vaccine, for instance, tetravalent DENV vaccine which is undergoing exploration in our lab. We have added an explanation in the revised manuscript (lines 342-345).

12. Line 562: should define FNT.

Response: We have defined "FNT" in the revised manuscript (line 750).

13. Line 588: change "teasing" to "passing"?

Response: We have corrected this error in the revised manuscript (lines 764 and 775).

14. Line 865, Fig 2a: Line 865 states IFNAR -/- mice were challenged 2 weeks after 2nd immunization. Figure 2a says they were challenged 3 weeks after immunization.

Response: We have rectified this inconsistency in the main text, the methods, the figures and the figure legends in the revised manuscript. All the ZIKV challenge assays regarding IFNAR$^{-/-}$ mice, except that shown in the revised Supplementary Fig. 6, were performed at 3 weeks after the final immunization, one week apart from the bleeding.

15. Figure 3A: The mating schedule is a little unclear from this image. Suggest changing the branch out point immediately after 1st immunization (before 2wks). Based on where the two lines branch out at the current scheme, it looks like two rounds of mating occurred in the 2-dose study line.

Response: We have re-drawn the schematic diagrams of animal experiments, as

shown in the revised Fig. 2a, 2h, 3a, 4a, to more clearly depict the timeline.

16. Sup Fig 1a and 9a: what is the meaning of "optional" indicated in both plasmid figures?

Response: The "optional" in the original Supplementary Fig. 1a and 9a indicated the presence of signal peptide in the antigen constructs while absence in the reporter constructs. We have deleted this confusing mark in the revised Supplementary Fig. 1a and 9a. Alternatively, we expounded this point in the figure legends.

17. Sup Fig 1h and 9e: It would be useful to provide a purpose of the junction site PCR and sequence validation.

Response: Following your valuable suggestion, we have revised the section regarding the junction site PCR and sequencing, in order to clarify that the purpose of this assay was to confirm the successful and precise circularization of the RNA constructs (lines 130-132 in the revised manuscript).

18. nAb was measured after immunization, but was never measured after virus challenging from the survivors. The post challenge nAb titers would be useful to determine whether the vaccine provided a solid sterile immunity (no boost effect of the post challenge nAb titer). It is somewhat surprising that it was not done for the vaccine study.

Response: Following your insightful suggestion, we have tested the nAb titers in the immunized IFNAR$^{-/-}$ mice at 7 days after challenge. This new data has been presented in the revised Supplementary Fig. 11i (lines 379-383 in the revised manuscript).

The post-challenge nAb titers increased approximately 2.8-fold in mice receiving 20 μg of prototype EN(LNP), and 4.2-fold in those receiving 5 μg of optimized EN(LNP). Interestingly, in mice receiving 20 μg of optimized EN(LNP), the increase of nAb titers was more moderate, at about 1.6-fold. These findings indicate that, although a single dose of optimized circRNA vaccine may not achieve sterilizing immunity, it effectively suppresses viral replication and spread following the challenge.

Reviewer #3 (Remarks to the Author):

In the manuscript entitled "A single dose circular RNA vaccine completely prevents Zika virus infection without enhancing dengue severity" by Liu et al., the authors developed ZIKV vaccines in a form of circular RNA encoding the envelop protein domain III and NS1 and reported the efficacy of the vaccine candidates using C57BL/6 and IFNAR-/- mouse strains. The study results showed that maternal immunity elicited by circRNA encoding dimeric EDIII, fused to human IgG1 Fc fragment (EDIII-Fc) were superior to circRNA encoding monomeric or trimeric EDII in terms of activation of germinal center reactions, facilitating high neutralizing antibody production and induction of CD4/CD8 T cells producing cytokines. One-day-old neonates born from dams immunized with either one- or two dose vaccines survived after lethal challenge but do not prevent viral infection and morbidity. CircRNA encoding either ZIKV EDIII-Fc or NS1 alone provides partial protection against ZIKV infection and developmental retardation. Further, optimized circRNA EN(LNP), which increases stability and protein translation, reduced viral RNA burden in the neonatal brains and increase protective efficacy that was durable up to 12 weeks. The authors concluded that two-dose immunizations of circRNA EN (LNP) provide full protection against ZIKV infection while avoiding ADE of DENV, but not by a single dose immunization. The results highlight that circRNA could be a platform for a safe and effective flavivirus vaccine.

Major comments:

Overall, the manuscript is well-written. The studies were well-designed with proper controls. The authors made thorough efforts to improve the efficacy of LNP encapsulated optimized circRNA vaccine encoding EDIII and NS1. The study results showed promising outcomes of protection of adult and neonate C57BL/6 and IFANR$^{-/-}$ mice against ZIKV infection while avoiding possible ADE of DENV, which is a significant risk factor of flavivirus infection. However, I have a few concerns regarding data presentation, and the conclusion in the discussion was overstated and not fully supported by the data presented in the current version of manuscript. I suggest that the

authors revise the conclusion to adhere to the results. See below the specific comment below.

Response: We appreciate your valuable comments and suggestions, which are important for improving the quality of our manuscript. We have revised the Figures to present the design and data more clearly and precisely, provided new data regarding the infection of mouse-adapted DENV2 in neonatal mice, and corrected the overstatements and misleading descriptions. We believe the revised manuscript has addressed all your concerns. Please see our point-by-point responses to your specific comments, as listed below.

Additional comments:

Figure 1f. Virus-binding antibody titers (d) were measured by one (ELISA) and neutralization antibody titers were measured by the other assay (FNT). Although the readouts of antibody response are reciprocal serum dilutions, it doesn't justify presenting the ratios of neutralizing antibodies to virus-binding antibodies (f). Virus-neutralizing antibodies are not exclusively IgG isotypes. IgA and IgM can be neutralizing yet virus-binding antibodies in this study measured IgG only not total antibodies. I doubt that the simple mathematical calculation of the ratio of serum titers is accurate and acceptable in the field. Without a thorough investigation using a method/platform to bridge two assays with well-established standard controls, the current form of data presentation can be misleading and inaccurate. If there are references supporting such data presentation, please add them or provide justifications. Otherwise, I suggest deleting Fig 1f. However, the authors may discuss the result and point out that EDIII-Fd induced robust anti-ZIKV antibody response but poorly neutralizing relative to antibody response induced by EDIII-Fc.

Response: We agree with your notion that nAbs are not exclusively IgG isotypes, and IgA and IgM can be neutralizing. We originally calculated the ratios of nAb titers to IgG titers because the concentration of IgG antibodies in the peripheral blood of adult mice is significantly higher than that of IgM (by about 10-fold) and IgA (by about 30-fold) (Klein-Schneegans AS, et al., J Autoimmun., 1989). In the context of vaccination

or natural ZIKV infection, IgM may be robustly induced within 2 weeks and contributes significantly to the neutralizing activities of immune sera (Calvert AE, et al., J Clin Microbiol., 2021; Singh T, et al., Cell, 2022). IgA can also be induced, although its titers in immune sera are generally lower than those of IgG (Li Y, et al., EBioMedicine, 2023). Two recent studies evaluated the ratio of binding (IgG) versus nAbs as an indicator of the quality of antibody responses against SARS-CoV-2 (Karron RA et al., JCI Insight, 2022; Qu L, et al., Cell, 2022).

Nonetheless, to avoid any misleading or inaccurate presentation, we decided to remove the original Fig. 1f. We also corrected the conclusion of this section as "suggesting that EDIII-Fc circRNA elicits higher titers of IgG and nAbs than EDIII-Fd and EDIII circRNAs" (lines 150-151 in the revised manuscript).

Lines 163-165: the authors stated that "…, including paralysis of limbs and tails, retardation of brain growth, meningeal thickening…." and presented brain mass and viral titers in tissues (Suppl. Fig3) but the evidence/data to support limb & tail paralysis and meningeal thickness are missing. According to histology in Figures 2f, 3f, and 4f, the immunized groups show less cellular infiltration in the brain tissue and possibly less necrosis than in the brain of the PBS group, in which dura membrane blebbing and cellular infiltration seem to be pronounced. It is difficult to know whether ZIKV induces changes in meningeal thickness in the representative image of each of the groups. No measurable data supporting meningeal thickening in the brain, nor the paralysis scores of limbs and tails were presented. Either provide the supporting data or revise.

Response: Regarding the limb & tail paralysis, we had recorded these symptoms as neurological scores and presented them in the original manuscript, actually. The neurological scores, shown in the revised Fig. 2d, 3e, 4e, 6f, and 7e, encompassed the cumulative scores reflecting paralysis in both the four limbs and the tail. To address any confusion, we have provided more extensive interpretations in the main text and figure legends in the revised manuscript (lines 189-190, 1129-1130).

Regarding the meningeal thickness and inflammation, we have noted that it was difficult to accurately measure the meningeal thickness using the obtained brain tissue

sections. Alternatively, we opted to score the severity of the lymphocyte infiltration in meninges and the cortex laminar necrosis, using a single-blinded manner. In the revised Fig. 2f, 3g, 4g, 6h, and 7h, we have marked the meningeal lymphocyte infiltration and cortex laminar necrosis in the representative images. We have also provided the pathological scores, encompassing the severity of meningeal inflammation and cortex laminar necrosis, to quantify the brain damage caused by ZIKV infection as well as protective effects of circRNA immunization, as shown in the revised Fig. 2g, 3h, 4h, 7i, and Supplementary Fig. 11e.

Line 207-208/Fig 2d, " EDIII-Fc… prevented ZIKV-caused growth delay and mortality, alleviated neurological disorders,…." For clarification, indicate the score of 14 is the maximum severity of limb and tail paralyses per mouse either in figure legend or in the main body.

Response: Following your valuable suggestion, we have indicated the maximum score of limb and tail paralyses per mouse is 14 in the figure legend (lines 1129-1130 in the revised manuscript).

In Fig 2, add a line with "C57BL/6" above the panel of Fig2b-e and a line with "IFNAR-/-" above the panel of g-j.

Response: In accordance with your suggestion, we have split the original Fig. 2a into the revised Fig. 2a and 2h, in order to depict the timeline of each mouse experiment more clearly. We also split the original Fig. 3 into the revised Fig. 3 and Fig. 4 to present the data of single-dose and two-dose immunization models separately. Similar changes have been applied to the original Fig. 5c to enhance the clarity. In addition, in the revised Supplementary Fig 3, 5 and 11, we have added lines and annotations to specify the animal models from which the data were obtained.

Line 232, clarify what do the authors meant by "more than half showed"; does this mean more than half of the mice within the EDIII-Fc group?

Response: Indeed, this phrase means more than half of the mice within the EDIII-Fc

group. We have refined this confusing rephrase, as "more than half of EDIII-Fc sera" (line 276 in the revised manuscript).

In Fig 5h and other figures (2f, 3f and 3h), the presentation of compact images side-by-side is difficult to identify the point of the interest other than the increase of cellular infiltration possibly also cellular necrosis, no localization of viral proteins- particularly to readers with no pathological background. Add a short description of the point of interest indicated with boxes, arrows, or arrowheads that would be helpful to understand the presence or absence of pathology caused by viral infection and protection, respectively.

Response: To address the concern regarding the difficulty of identifying points of interest, we have marked the meningeal lymphocyte infiltration and cortex laminar necrosis in the representative images, as shown in the revised Fig. 2f, 3g, 4g, 6h, and 7h. In addition, we have conducted pathological assessments, with the results presented in the revised Fig. 2g, 3h, 4h, 7i, and Supplementary Fig. 11e. Statistical analysis confirms that immunization with circRNA vaccines significantly mitigates the brain damage caused by ZIKV infection, underscoring the protective efficacy of our approach.

To make it easy to follow, add a line with the indication of " IFNAR-/-" below Fig 5h and above the panel of i-k in order to separate data from C57BL/6 mice and IFNAR-/- mice.

Response: We have split the original Fig. 5c into the revised Fig. 6c and 6i to separately present the data from C57BL/6 mice and IFNAR$^{-/-}$ mice. We think this will clearly differentiate the animal models used.

Line 304/ Fig6 e: At 15 dpi, 2 out of 13 pups (15%) after optEN(LNP) and 6 out of 7 pups (86%) after 20ug pEN(LNP) still showed substantially high levels (~10^8 copies) of viral RNA (Fig6e) with measurable neurological score (Fig 6d). Although the results presented "promising potential" of Opt EN(LNP) but did not support full nor complete protection against ZIKV infection at 13 weeks post-vaccination. The author shouldn't

ignore these unprotected outliers and may discuss the result.

Response: We acknowledge that a single dose of EN(LNP), even with optimization, cannot confer complete protection in the neonates challenged at 11 weeks after maternal immunization. This outcome can be attributed to the diminished levels of nAbs and anti-NS1 antibodies at this time point, both of which are pivotal for the protection. In the revised manuscript, we described the short-term and long-term protective effects in two separate sections. The titles of these sections have been revised as "A single dose of optimized circRNA vaccine confers potent protection" and "Protective immunity elicited by optimized circRNA vaccine is durable" (lines 317 and 386 in the revised manuscript, respectively). We believe that these amendments more precisely summarize the key findings in each respective section.

Line 307-8, the authors state that "Optimized circRNA vaccine has the potential to trigger…", which is likely, but data present the duration of protective immunity up to 12 weeks, which is relatively short. Consider revising "Optimized circRNA vaccine induced protective immunity lasting at least up to 12 weeks" to adhere the results presented.

Response: Following your insightful suggestion, we have revised this statement as "Optimized circRNA vaccine, therefore, induces protective immunity lasting at least up to 11 weeks in mice" to adhere the results shown in the revised Fig. 7 (lines 401-402 in the revised manuscript).

Line 314, The statement "….the combination of EDIII-Fc and NS1 circRNAs confers complete protection in two mouse models (Fig. 3, 5)." is not correct. In Figure 3e, 3 out of 7 C57BL/6 mice (41%) after single dose immunization with EN(LNP) showed substantially high levels (108 copies/gram) of viral RNA. Only after two-dose immunizations with EN(LNP), mice were protected against infection but not after EN(RNA) immunization. In IFNAR-/-, viral RNA was also detected in all mice after single dose immunization (suppl. Fig 6g). I agree that the results presented are promising but, in the discussion, the conclusion remarks are overstated. Revise the

conclusive statement adhering to the results presented in this manuscript.

Response: To avoid the overstatement as you mentioned, we have replaced the "complete" with "effective" in the revised manuscript (line 435).

Line 328, "our circRNA vaccine is quite promising …" sounds colloquial. I suggest revising "our circRNA vaccine produces promising results…"

Response: We have revised this sentence to avoid the colloquial expression, following your kind suggestion (line 450 in the revised manuscript).

Line 356, ".., but barely lymphocytes, can be vertically…" "barely" is not correct. In healthy conditions, maternal lymphocytes are not on the fetal side. Please revise.

Response: We have revised this incorrect description as "but not lymphocytes" (line 488 in the revised manuscript).

Line 383-392, the authors described that the mouse adaptive DENV2 generation from the parental 16681 strain was obtained after 3 passages in the neonatal brain. This is relatively low passages compared with the mouse adaption achieved 10 passages of a parental strain, PL0468 in AG129 mice (DOI: 10.1128/JVI.00062-06). Relative to the old method, 3 passages in B6 neonates seem to be a huge improvement. Provide GenBank accession no. of the mouse-adaptive DENV. In addition to sequence analysis for identification of site mutations (supplementary Figure 8a), the authors should provide supplementary results showing productive DENV infection in immunocompetent or IFNAR-/- mice. If the characterization and pathogenesis of the current mouse-adapted DENV 2 were already published, please provide references. The citation 46 alone is insufficient.

Response: Please see also our response to Reviewer 1's comment 8.

We have submitted the genomic sequences of the mouse-adapted DENV2 in GeneBank (No. PQ008452).

We generated mouse-adapted DENV2 from DENV2 strain 16681 (GenBank No. U87411), through alternatively passaging in 1-day-old C57BL/6 mice and Vero cells

for 3 cycles. It has been reported that a subcutaneous injection with DENV2 (NGC strain) led to lethal brain infection in 1-day-old C57BL/6 mice (https://www.fda.gov/media/168791/download). Another study showed that an intranasal inoculation with DENV2 also led to severe brain infection in 3-day-old suckling BALB/c mice (Qiu M et al., Viruses, 2022). Live DENV-2 virions could be isolated from the brain tissues and passaged on Vero cells (Qiu M et al., Viruses, 2022). The rapid occurrence of the mouse-adapted variant in our model might be attributed to the following reasons: i. The genome of DENV2 appeared to be less stable in Vero cells than in C6/36 cells (Chen WJ, et al., Intervirology, 2003), leading to a variety of quasi-species populations that can be selected in infected mice; ii. The higher culture temperature of Vero cells (37 ℃) relative to C6/36 cells (usually 28 ℃) maybe also contribute to the occurrence of quasi-species population harboring virulence-imparting mutations (Modak A, et al., FASEB J, 2023); iii. The successful establishment of infection by DENV2 in neonatal brains need to break through more barriers in comparison with the method adopted by Orozco S et al (intravascular inoculation followed by viral isolation from the sera) (Orozco S et al., J Gen Virol, 2012; Shresta S et al., J Virol, 2006), thereby exerting stronger selective pressure.

We identified three amino acid substitutions in the mouse-adapted variant compared to the parental DENV2: NS1 K174N, NS2A L181V, and NS5 A196T. Although the exact contribution of each mutation to the virulence remains to be determined, an intracerebral injection of the adapted variant led to faster mortality in 1-day-old ICR mice in comparison with the parental DENV2 (revised Supplementary Fig. 8b, c). We have added this result to the revised manuscript (lines 292-298).

Line 562, spell out "FNT" as it appears the first time in the main body of the manuscript.

Response: We have spelled out "FNT" (line 750 in the revised manuscript), following your kind suggestion.

Line 641, provide the source/vendor of DENV monoclonal antibody clone 8-4, I am not sure if the clone is correct.

Response: The monoclonal antibody (mAb) in question was originally isolated from a ZIKV-infected patient and was found to have binding activities to the EDI/II of both ZIKV and DENV (Yu L et al., JCI Insight, 2017). It was designated as mAb ZK8-4 in the literature. To maintain consistency with the existing literature and avoid confusion, we retained the original clone designation mAb ZK8-4 and provided a brief introduction of this mAb (line 836 in the revised manuscript).

Reviewer #4 (Remarks to the Author):

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Response: We appreciate your valuable review work. We believe that the quality of this manuscript will be substantially improved by addressing your comments and concerns.

Reviewer #1 (Remarks to the Author):

The authors have addressed all the major concerns raised on their original manuscript. This revised version now reads well and provides a balanced interpretation of the data.

Response: We greatly appreciate your efforts and suggestions, which significantly helped us to improve the quality of this manuscript.

Reviewer #2 (Remarks to the Author):

We (primary reviewer and co-reviewer) thank the authors for the thorough responses including new experiment data and revision of the manuscript. All our concerns have been adequately addressed, and the revised manuscript has been greatly improved in clarity and provides significant findings in developing and optimizing a circular RNA ZIKV vaccine. Below are 2 minor comments of the revised manuscript:

Response: We greatly appreciate your valuable suggestions, which are significantly helpful for improving the quality of our manuscript.

1. Fig 8a and b: Missing the ZIKV sera data in Fig 8a or b. Since the ZIKV sera was used in Fig 8c as the in vitro ADE positive control, it would be better to include its overall IgG titers to various viral E antigens in Fig 8a and/or 8b (as shown in Fig 5a).

Response: We did not included the data regarding ZIKV sera into Fig 8a and b because the cross-reactivity of ZIKV sera, collected at 2 to 12 weeks after infection, have been presented in Supplementary Fig. 7. The inclusion of these data here is somewhat redundant. Please note that the ZIKV sera used in Fig. 8c and 8d were derived from the same mice as those in Supplementary Fig. 7.

2. Figure legends for all brain H&E staining images: Shouldn't black arrows indicate cortex laminar necrosis, and cyan blue arrows lymphocyte infiltration? All legends for these figures indicated opposite.

Response: Actually, black arrows indicate lymphocyte infiltration of meninges, whereas cyan arrows indicate necrotic cells in cerebral cortex. We have revised the relevant legends to more clearly describe this point, as "Black arrows, meningeal lymphocyte infiltration. Cyan arrows, necrotic cells in cortex." Similar pathological

observations in the context of ZIKV infection have been reported (Dowall SD et al., PLoS Negl Trop Dis., 2016)

Reviewer #3 (Remarks to the Author):

The authors have addressed the reviewers' comments made on the previous submission. Additionally, the authors have made an extensive revision that improved clarity and provided additional supporting information. The results presented in this manuscript strongly support the conclusions.

Response: We greatly appreciate your valuable suggestions. We have revised all the incorrect or misleading words or sentences, following your kind suggestions, as listed below.

I have a few minor suggestions and corrections for citations as indicated below:

line 159, " .. plasma cells were significantly detected…" sounds awkward and I suggest revising 'significantly high' or '..significantly higher… than in the mice immunized with EDIII-Fd or the others'.

Response: We have revised this sentence as "plasma cells were significantly higher in EDIII-Fc-immunized mice than in the mice immunized with EDIII-Fd or EDIII", following your kind suggestion.

line 167, instead of "but", use either 'and' or 'while these were…'

Response: We have replaced "but" with "and", following your kind suggestion.

lines 297 and 310, in the description of passive ZIKV-immune sera transfer, there is a discrepancy in the volume of serum samples, "20 uL" on line 297 ' 200 uL of diluted serum" on line 310 and Figure legend 5C. Please be consistent.

Response: We have described this point more clearly with "200 μl tenfold diluted ZIKV sera or immune sera" in line 298 in the revised manuscript.

line 449, 'favorable' may be better than "promising"

Response: We have replaced "promising" with "favorable", following your kind

suggestion.

line 547, Vero cells is CCL-81 not "NCCL-81"

Response: We have corrected this typo in the revised manuscript.

line 562, "1x10^4 FFUs of DENV2", delete 's' in FFUs

Response: we have deleted “s” in FFUs herein and throughout the manuscript.

line 813, change from "qRT-qPCR" to "qRT-PCR" consistent with line 812

Response: We have corrected this typo in the revised manuscript.

Comments on citations:

line 52, Amaya et al., 2023 is inappropriate. Please correct.

Response: we have replaced this citation with “https://www.who.int/publications/m/item/zika-epidemiology-update-may-2024” in the revised manuscript. We also updated the number of countries with current or previous Zika virus transmission.

line 515, the statement “…. the pregnancy mouse models used in other studies 61, 62” including citation 62 is incorrect. Reference 62 used a litter-bearing non-human primate, marmoset pregnancy model not a mouse model. Authors may replace reference 62 with another immunocompetent mouse pregnancy model, human Stat2 knock-in (hSTAT2KI) mouse (PMID: 38360793). hSTAT2KI mouse model recapitulates vertical transmission of ZIKV during pregnancy while ZIKV causes fetal demise in pregnant C57BL/6 mouse model (reference 61). Alternatively, authors may revise the statement from “…the pregnancy mouse models…” to “the preclinical pregnancy models” referring mouse and marmoset models.

Response: We have revised the statement from “the pregnancy mouse models” to “the preclinical pregnancy models”. We also added this reference (PMID: 38360793) in the revised manuscript.

Reviewer #4 (Remarks to the Author):

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Response: We greatly appreciate your valuable suggestions, which are significantly helpful for improving the quality of our manuscript.